Allocating Decision Rights: A Public-Goods Experiment on (Un)Democratic Reversal

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Abstract

Why do citizens authorize weakly constrained power, and do they reclaim decision rights when the crisis that justified it recedes? This project tests a demand-side microfoundation of democratic backsliding in a one-session public-goods laboratory experiment with 400 participants in Finland. Across twenty rounds, anonymously rematched five-person groups vote between citizen-led provision and one-person rule. During the crisis, citizen-led provision is less productive; leader control produces more, compels equal allocations, and lets the leader move fund points to their payoff. After Block 1, ten-person pools are randomly assigned to recovery or persistence. Under recovery, public-service conditions improve and citizen-led productivity rises to match leader-led productivity; under persistence, the crisis and productivity gap remain. Leader powers remain fixed. Among participants who supported leader control in Block 1's final round, the randomized contrast estimates whether recovery causes a return to citizen control. The first and final Block-2 ballots distinguish immediate from sustained reversal.

Experiment prototype (production phase):
`https://panel-lab-experiment.onrender.com`

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1 Research Question and Contribution

Democratic backsliding depends not only on leaders who weaken constraints but also on citizens who authorize them. An institutional crisis—such as crime, immigration, or inequality—can make that authorization attractive by giving concentrated power a practical advantage. This project asks what happens when that advantage *disappears*: *once the crisis is over, do the same citizens who supported illiberalism reclaim their rights?* This within-person return to citizen control is the **(*un*)democratic reversal** the project seeks to explain.

We build on the democratic backsliding and public-good games literatures. First, the democratic backsliding literature has traced the forward movement from threat to weaker democratic commitments. Perceived terrorist threat can make even less-authoritarian citizens more willing to restrict civil liberties, while immigration-threat frames can weaken liberal-democratic citizenship norms (Hetherington and Suhay 2011; Goodman 2021). In Finland itself, citizens sometimes condone democratic transgressions when the offending leader offers ideological or policy benefits (Saikkonen and Christensen 2023).

Second, the public-goods experimental literature supplies the economic logic behind that choice. In a standard voluntary-contribution game, keeping one's own endowment is individually optimal even though full provision maximizes group earnings. With repeated play, imperfect conditional cooperation can drive initially positive contributions downward (Fehr and Gächter 2000, pp. 982–984; Fischbacher and Gächter 2010, pp. 541–554). Punishment and formal enforcement can arrest that decline and, with enough experience, generate net gains (Fehr and Gächter 2000, pp. 993–994; Gächter, Renner, and Sefton 2008, p. 1510). Participants also form institutions and elect delegates (Kosfeld, Okada, and Riedl 2009; Hamman, Weber, and Woon 2011), while repeated experience can draw them toward a sanctioning environment once it performs better (Gürerk, Irlenbusch, and Rockenbach 2006, pp. 108–110). Leader control in this experiment thus offers a genuine economic solution to a familiar collective-action problem.

The public-good games literature also explains why endogenous institutional choice belongs inside the experiment. Dal Bó, Foster, and Putterman (2010, p. 2205) show that democratically selected rules can affect subsequent cooperation differently from otherwise identical imposed rules. Sutter, Haigner, and Kocher (2010, p. 1540) and Markussen, Putterman, and Tyran (2014, pp. 321–322) likewise find that voting over public-good institutions changes their performance. Classic evidence further shows that communication and self-chosen enforcement can sustain collective action without an external authority (Ostrom, Walker, and Gardner 1992, pp. 412–414). Together, these findings establish why authority can be attractive and why institutional choice can shape public-good provision.

Read together, the two literatures explain why citizens may authorize concentrated power during a crisis, but they do not follow what happens when the crisis-based benefit recedes. This project takes that next step. Its specific contribution is to test **whether citizens withdraw an earlier authorization after the structural condition that supported it changes**.

To test this argument, the project uses an incentivized public-goods game conducted over repeated rounds in a single session. Participants vote between two rules for governing the same shared fund. If a simple majority chooses the citizen-led rule (D), each person controls their own points and no participant can remove points from the fund. If a simple majority chooses the leader-led rule (A), one randomly appointed person sets the same allocation for everyone and may move some fund points to their own payoff. A vote for A therefore authorizes "weakly constrained" centralized control: one participant replaces five individual allocation decisions with a binding group decision and gains discretion over part of the shared fund.

The experiment then manipulates the setting in which participants make this choice. Everyone begins with the same public-service crisis: "`Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled.`" During the first ten rounds, the leader-led rule produces more points and requires the same allocation from everyone, effectively minimizing free riding. Afterward, each ten-person pool is randomly assigned to **institutional recovery** or **institutional persistence**. Under recovery, the budget passes, waiting times begin to fall, planning resumes, and the citizen-led payoff rate rises to match the leader-led rate. Under persistence, the crisis and original rate gap remain. Because the leader's powers and rules do not change, comparing later choices identifies whether recovery causes previous supporters to return to citizen control.

The incentive structure makes this institutional choice consequential; **the *allocation of decision rights* makes it a choice about whether authority should remain democratically constrained**. A citizen who returns to the citizen-led rule withdraws an earlier authorization that allowed one person to replace five individual decisions, impose a binding allocation, and exercise discretion over the shared fund. Because leader control still compels equal allocations and prevents unilateral free-riding after recovery, returning to citizen control reclaims individual choice while accepting the collective-action problem that comes with it. This within-person withdrawal is the demand-side act at the heart of (un)democratic reversal: authority concentrated during a crisis is returned to citizens when its structural justification recedes.

In simple terms

In each round, five citizens vote on how to manage a shared fund. Under "Each person chooses" (D), each controls their own allocation. Under "One person chooses for the group" (A), the software appoints a citizen from among those who have served least often. That leader sets the same allocation for everyone and may move fund points to their own payoff. Citizens see the result, including any such transfer, after every round.

In Block 1, each fund point produces 1.50 points under D and 2.50 under A, divided equally among five citizens.

For example, 10 points placed in the shared fund produce $10 \times 1.5 = 15$ under D or $10 \times 2.5 = 25$ under A, giving each citizen $\frac{15}{5} = 3$ or $\frac{25}{5} = 5$ points. After Block 1, each ten-person pool is randomly assigned to institutional recovery, under which public-service conditions improve and the D rate rises from 1.50 to 2.50, or institutional persistence, under which the crisis and 1.50 rate remain. The A rate and powers do not change.

Citizens then vote again in Block 2. Among previous A supporters, returning to D after recovery means reclaiming individual decision rights once conditions improve and one-person control loses its one-point productivity premium: both methods now use the 2.50 rate. Continuing with A means retaining that authority after its crisis advantage disappears. The causal contrast compares how often these citizens return to D under recovery with how often they return under persistence. Because the same citizen cannot face both conditions, persistence provides the counterfactual: **how often comparable citizens would have returned to D had the crisis and productivity premium remained**. **Random assignment therefore identifies how much institutional recovery changes (un)democratic reversal**.

2 Theory

2.1 Crisis and the productivity premium

A crisis can change institutional support by changing the relative performance of available rules. When ordinary collective procedures deliver poor outcomes, a weakly constrained leader can gain support by promising faster or more effective provision (Kingzette et al. 2021; Saikkonen and Christensen 2023; Wunsch, Jacob, and Derksen 2025). The experiment implements that argument through a public-service scenario backed by a consequential production technology. Recovery therefore changes both the stated conditions and the payoff rate that makes those conditions matter inside the game.

In Block 1, the citizen-led fund (D) turns each fund point into 1.50 "public-service points." The leader-led fund (A) turns it into 2.50 points and forces an equal allocation from all five citizens. Leader control therefore combines a productivity premium with compulsory coordination. It also lets one participant decide for the group and move fund points to a personal account. Repeated public-good experiments show why the first two features are consequential: voluntary provision is vulnerable to declining cooperation, whereas enforcement can stabilize it (Fehr and Gächter 2000, pp. 982–984, 993–994; Fischbacher and Gächter 2010, pp. 541–554).

The transfer option under the A rule gives "weak constraints" a concrete meaning. By choosing A, citizens gain coordinated provision but also authorize the leader to redirect shared points; **an actual transfer realizes the cost of that discretion**. Later ballots reveal whether citizens continue accepting concentrated power after seeing how it is used or instead reclaim individual control. The experimental design therefore captures citizens' *tolerance of weak constraints* (not the prevalence of, e.g., corruption, in Finland.)

H1: Crisis authorization. During Block 1, support for the leader-led method (A) will rise when it delivers more productive and coordinated provision, but fall when leaders move shared-fund points to their own payoffs.

2.2 Citizen authorization and democratic backsliding

Leaders carry out democratic backsliding, but citizens can help make it possible. The experimental design isolates that demand-side step by separating the *authorization* of weakly constrained authority from its *exercise*. That is, the ballot records whether a citizen authorizes A; the transfer (ρ) records how the (randomly) selected leader uses the resulting discretion (i.e., if they move shared-fund points to their own payoffs).

H2: Exercise of discretion. Moving one fund point to a personal account raises the leader's payoff by 0.50 points while lowering every other citizen's payoff by 0.50 points. Leaders who place greater weight on their own earnings than on group earnings will therefore make larger transfers, whereas leaders who value group outcomes will leave more points in the fund. Among citizens who observe these choices, larger transfers will be followed by lower support for A in the next round.

2.3 Recovery and democratic reversal

Each game requires five citizens, so three votes produce a clear majority. A fixed group of five, however, could develop stable reputations or patterns of retaliation across rounds. The design therefore keeps participants in a ten-person matching pool and reshuffles them into two temporary groups of five before each round (see Figure 2). This is the smallest pool that allows partners to change. The pool, denoted by g, receives the Block-2 condition as a whole: let $T_g = 1$ denote institutional recovery and $T_g = 0$ institutional persistence. Let m_{gb} denote the citizen-led fund's multiplier in pool g and block b. The leader's productivity premium is

$$\delta_{gb} = 2.50 - m_{gb} \tag{1}$$

where δ_{gb} is the productivity premium of the leader-led fund over the citizen-led fund. Everyone faces the crisis and $\delta_{g1} = 1$ in Block 1. In Block 2, institutional recovery reports that public-service conditions have improved and sets $\delta_{g2} = 0$, whereas persistence reports that the crisis continues and keeps $\delta_{g2} = 1$. The leader's compulsory allocation, transfer option, selection rule, and 2.50 fund rate do not change. For example, in Block 1 each fund point produces 1.50 points under D and 2.50 under A, so $\delta_{g1} = 2.50 - 1.50 = 1$: leader control produces one additional public-service point. Under recovery, the D rate rises to 2.50, so $\delta_{g2} = 2.50 - 2.50 = 0$ and the two methods are equally productive. Under persistence, the D rate remains 1.50 and the one-point premium remains.

Among participants who chose A at the end of Block 1, the analysis compares how many return to D in institutional recovery pools ($T_g = 1, \delta_{g2} = 0$) with how many return in institutional persistence pools ($T_g = 0, \delta_{g2} = 1$). **This difference is the causal effect of institutional recovery on (un)democratic reversal** (see Figure 5); eliminating the leader's productivity premium is its material implementation in the game. Institutional persistence provides the counterfactual: *how often citizens would have returned to D if the crisis and premium had remained.*

H3: Immediate structural response. Among participants who choose A in Block 1's final round, assignment to institutional recovery will increase the probability of choosing D in Block 2's first round.

H4: Sustained democratic reversal. Among participants who choose A in Block 1's final round, assignment to institutional recovery will increase the probability of choosing D in Block 2's final round. The recovery-versus-persistence contrast identifies the effect of recovery relative to a continuing crisis and productivity gap.

3 Experimental Design

3.1 Overview

Each laboratory cohort completes one 75–90 minute session with two ten-round blocks. Participants remain in ten-person matching pools but are reshuffled into two anonymous five-person groups before every paid round. After Block 1, each entire pool is assigned to recovery or persistence, learns whether public-service conditions improved or remained strained, and sees its Block-2 fund rates. Participants therefore interact repeatedly without forming stable five-person teams. In the figures, p_i identifies participant i.

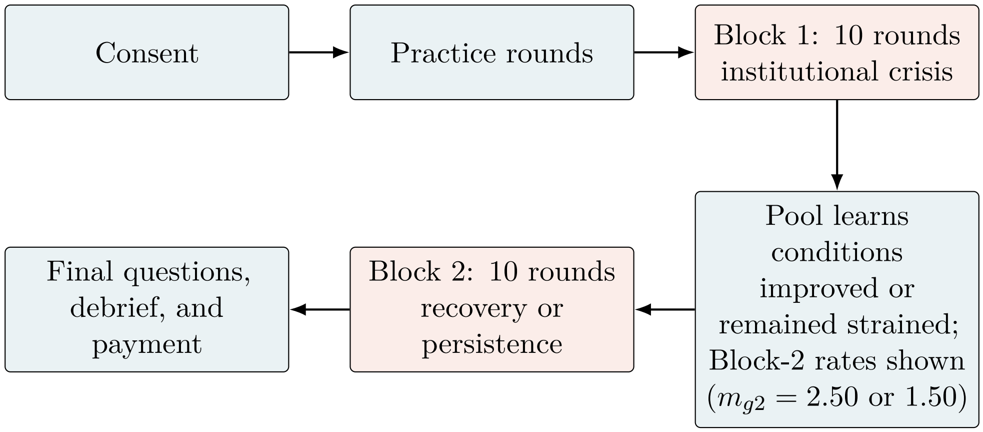


Figure 1: One-session sequence. Each paid round includes an individual choice, allocation, feedback, and anonymous rematching.

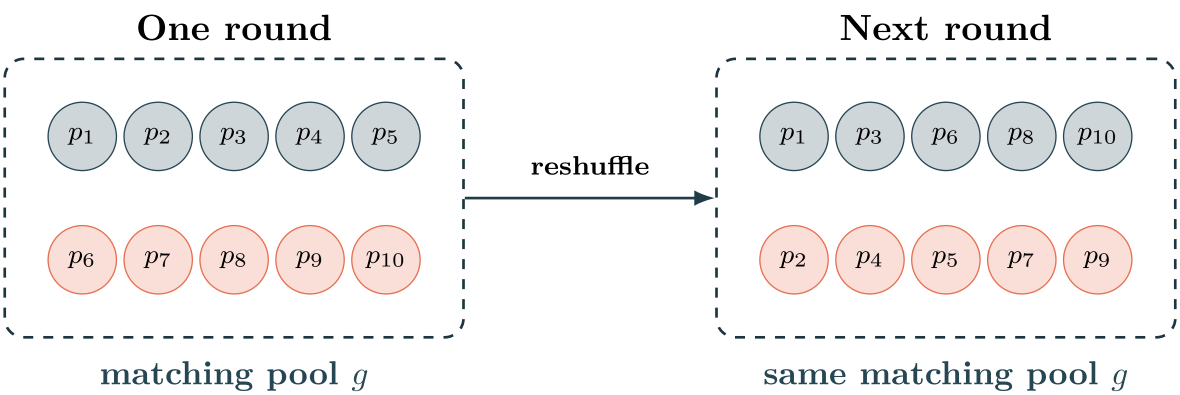


Figure 2: Anonymous rematching within a ten-person pool.

Notes: Each participant p_i remains in pool g. Colors mark temporary five-person groups; the software reshuffles them within the pool before every paid round. Recovery or persistence applies to the whole pool in Block 2.

In simple terms

The ten-person pool permits interaction without fixing the same five-person team. Round-by-round reshuffling limits stable reputations, collusion, and retaliation, while keeping participants inside one treatment environment. Reshuffling everyone would either mix treatment environments or require session-level assignment, leaving far fewer independent randomized units. Reversal is measured for individuals, but treatment (T_g) is assigned to pools. Statistical uncertainty is therefore clustered by pool; temporary groups organize play but are not treatment units.

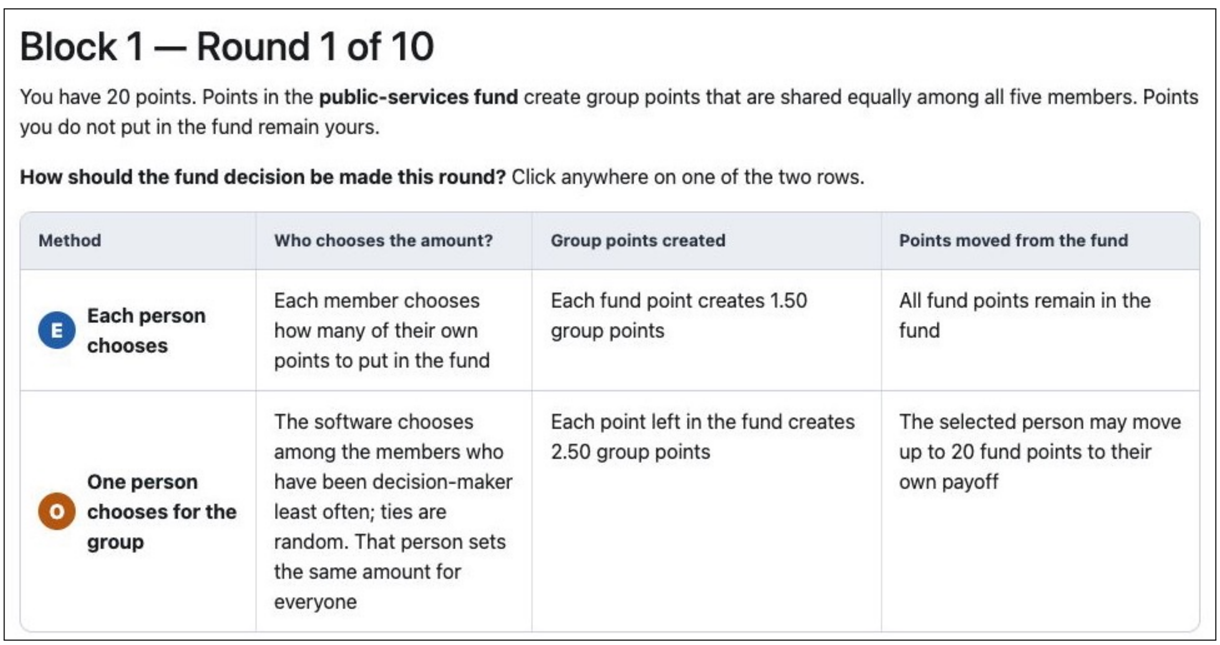
3.2 The institutional-choice public-good game

Each of the five citizens starts a round with 20 points and privately chooses between the citizen-led method (D) and the leader-led method (A). Let $V_{i,b,r} \in \{\mathsf{D}, \mathsf{A}\}$ denote citizen i's ballot in block b and round r. At least three votes make a method binding for that round (simple majority). The group sees the vote totals and the selected rule, but not individual ballots. On screen, the two options carry the neutral labels "Each person chooses" and "One person chooses for the group" (see Figure 3).

Table 1: Implemented design parameters

Feature	Implemented value
Laboratory session	One synchronized visit, approximately 75–90 minutes
Paid rounds	10 per block (20 total), preceded by unpaid practice
Group and matching pool	5-person groups; stable 10-person matching pools
Rematching	Anonymous rematching within pool before every paid round
Round endowment	20 points per participant
Citizen-led multiplier	1.50 in Block 1; 2.50 recovery / 1.50 persistence in Block 2
Leader-led multiplier	2.50 in both blocks and conditions
Leader's personal transfer	0–20 fund points, limited by the number of points in the fund
Performance payment	One randomly selected game round from each block
Exchange rate	EUR 0.20 per point
Show-up payment	EUR 10.00

The random-round rule limits wealth effects, hedging across rounds, and excessive payment while preserving incentives in every decision.


Block 1 — Round 1 of 10

You have 20 points. Points in the **public-services fund** create group points that are shared equally among all five members. Points you do not put in the fund remain yours.

How should the fund decision be made this round? Click anywhere on one of the two rows.

Method	Who chooses the amount?	Group points created	Points moved from the fund
E **Each person chooses**	Each member chooses how many of their own points to put in the fund	Each fund point creates 1.50 group points	All fund points remain in the fund
O **One person chooses for the group**	The software chooses among the members who have been decision-maker least often; ties are random. That person sets the same amount for everyone	Each point left in the fund creates 2.50 group points	The selected person may move up to 20 fund points to their own payoff

Figure 3: Participant screen for choosing the round's decision method. The fund rates update with the block and treatment condition.

Citizen-led procedure (D). Citizen i places $c_{igbr} \in \{0, \ldots, 20\}$ points in the public-services fund in round r and keeps the rest; j indexes the five citizens in their current group. Their payoff is

$$\pi^D_{igbr} = 20 - c_{igbr} + \frac{m_{gb}}{5} \sum_{j=1}^{5} c_{jgbr}. \tag{2}$$

One contributed point costs its contributor more than it returns privately, but benefits the group as a whole. Here, $m_{g1} = 1.50$; in Block 2, $m_{g2} = 2.50$ under recovery and 1.50 under persistence.

In simple terms

This is the **public-goods dilemma**. As an illustration under D, suppose citizen i contributes $c_{igbr} = 1$ and the other four free-ride, contributing $c_{jgbr} = 0$ for $j \neq i$. Because every citizen starts the round with 20 points, citizen i keeps $20 - 1 = 19$ after contributing one point; each free-rider keeps 20. During the crisis, everyone receives $\frac{m_{g1}}{5} = \frac{1.50}{5} = 0.30$ from i's point. **Hence, i earns $\pi^D_{igbr} = 19.30$, while each free-rider earns $\pi^D_{jgbr} = 20.30$.** Persistence ($T_g = 0$) gives the same payoffs. Under recovery ($T_g = 1$), the return is $\frac{m_{g2}}{5} = \frac{2.50}{5} = 0.50$, **yielding 19.50 for i and 20.50 for each free-rider**.
Zero is therefore feasible and strategically meaningful under D: individually, free-riding pays; collectively, provision pays. That dilemma makes A appealing because one-person control can impose equal allocations. Reversal is the return to D after recovery removes that productivity premium, despite D's continued exposure to free-riding.

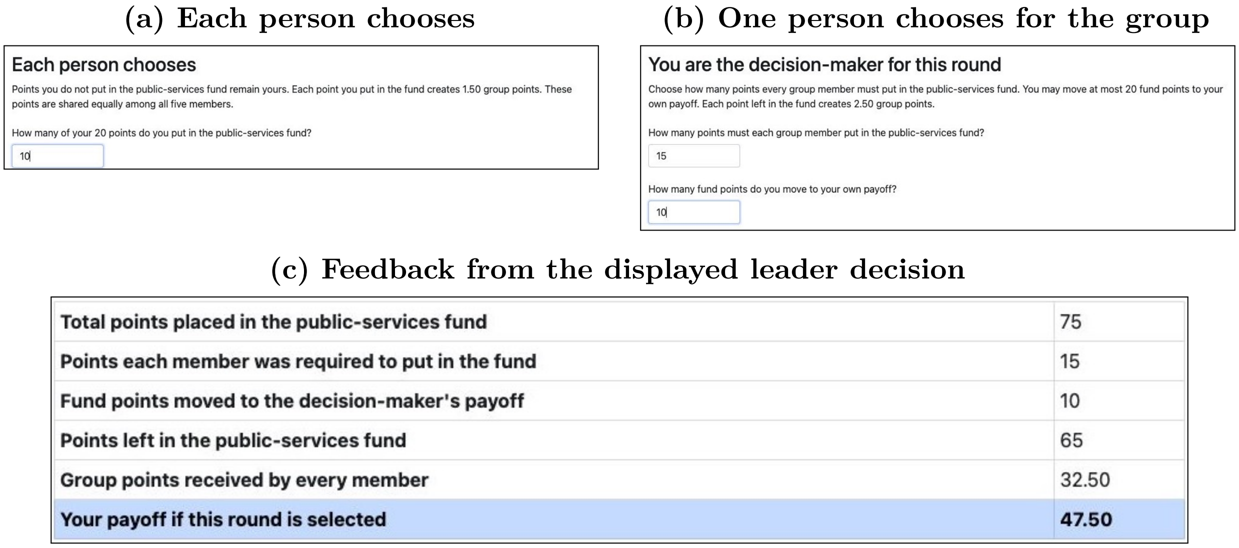
Leader-led procedure (A). The software selects a leader from the members who have served least often, breaking ties at random. The leader sets one required allocation $t \in \{0, \ldots, 20\}$ for all five citizens and may move $\rho \in \{0, \ldots, 20\}$ fund points to a personal account, subject to $\rho \leq 5t$. The remaining fund $5t - \rho$ always uses multiplier 2.50. A nonleader earns

$$\pi^A_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho), \tag{3}$$

while the leader earns

$$\pi^{A,L}_{igbr} = 20 - t + \frac{2.50}{5}(5t - \rho) + \rho. \tag{4}$$

Everyone observes the required allocation, transfer, public-service return, and own payoff. Leader control coordinates provision and is more productive in Block 1, but it places the shared fund in another participant's hands.



Total points placed in the public-services fund	75
Points each member was required to put in the fund	15
Fund points moved to the decision-maker's payoff	10
Points left in the public-services fund	65
Group points received by every member	32.50
Your payoff if this round is selected	**47.50**

Figure 4: Method-specific decisions and round feedback. The entered amounts illustrate the interface and are not participant data.

How incentives map onto reversal. During Block 1, every pool faces the same crisis incentives: each point placed in the citizen-led fund produces 1.50 group points ($m_{g1} = 1.50$), whereas each point left in the leader-led fund produces 2.50. Leader control therefore offers a one-point productivity premium and can also prevent unilateral free-riding by imposing the same allocation on everyone. After Block 1, random assignment creates two paths. Under institutional recovery, public-service conditions improve and the citizen-led rate rises to $m_{g2} = 2.50$; under persistence, the crisis continues and the rate remains $m_{g2} = 1.50$. In both conditions, the leader-led rate, compulsory allocation, transfer option, and selection rule remain unchanged. Recovery thus makes fund points equally productive under the two methods without making their decision rules identical or guaranteeing equal payoffs. Accordingly, comparing previous A supporters across recovery and persistence identifies whether institutional recovery causes them to reclaim individual decision rights; persistence shows how often they would have returned through repeated play alone.

From institutional choice to (un)democratic reversal. The payoffs make institutional choice consequential; **the *allocation of decision rights* makes it a choice about whether authority remains democratically constrained**. A citizen who switches from A to D first authorizes one person to replace five individual decisions, impose a binding allocation, and exercise discretion over the shared fund, then withdraws that authorization after its crisis-based productivity advantage disappears. Because A still compels equal allocations and prevents unilateral free-riding after recovery, returning to D reclaims individual control while accepting the collective-action problem that comes with it. This within-person withdrawal is the demand-side act at the heart of (un)democratic reversal: authority concentrated during a crisis is returned to citizens when its structural justification recedes. A higher rate of A-to-D switching under recovery therefore shows that structural change can cause citizens to withdraw the support that sustains weakly constrained rule. *The experiment thus observes reversal as a consequential institutional act rather than a declaration of democratic sentiment.*

A simple payoff comparison

Suppose each citizen places 10 of 20 points in the fund under D. Under A, suppose the leader requires the same 10-point allocation and makes no personal transfer. Holding these moderate decisions fixed gives:

	Citizen control (D)	Leader control (A)
Crisis or persistence	25 points each	35 points each
Recovery	35 points each	35 points each

Recovery brings D up to A's production rate; it does not make D pay more in this benchmark. Actual payoffs still depend on citizens' allocations and the leader's transfer. **The behavioral question is therefore: *when citizen control becomes equally productive, does support shift away from leader control?***

Participants report their expected payoff under each method and the transfer they expect the leader to make at three points: before Block 1, Round 10, which records expectations before treatment; before Block 2, Round 1, which records their immediate expectations after learning their assigned condition; and before Block 2, Round 10, which records expectations after experiencing the new environment. Their clearest function is as a **manipulation and process check**: *did participants understand how recovery changed the relative performance of the two methods?* These reports help interpret changes in voting.

In simple terms

Five citizens receive 20 points and cast private ballots ($V_{i,b,r}$). Under "Each person chooses" (D), each citizen chooses a fund allocation (c_{igbr}). The total is multiplied by the current rate (m_{gb}), shared equally, and unallocated points remain personal.
Under "One person chooses for the group" (A), the software selects a decision-maker from those who have served least often. This person sets everyone's allocation (t) and may move fund points to their own payoff ($\rho \leq \min\{20, 5t\}$). The rest ($5t - \rho$) is multiplied by 2.50 and shared equally. Everyone sees the decision and own payoff (π^D_{igbr} or π^A_{igbr}) before rematching within pool g.

3.3 Block 1: crisis and leader authorization

After consent, participants receive short, neutral instructions, practice both methods without payment, and answer three comprehension questions. The interface never calls either method democratic or undemocratic.

Block 1 begins with the following crisis: "`Parliament has failed to pass a budget for eight months, healthcare waiting times have doubled, and planning for public services has stalled.`" Each fund point then produces 1.50 under D and 2.50 under A. The selected decision-maker under A may move up to 20 fund points to their own payoff.

The ten paid rounds repeat the same sequence: private choice, majority selection, the fund decision, feedback, and rematching. Expected payoffs and transfers are elicited before the final choice. Questions about democratic constraints appear only after both paid blocks. The software records all choices and payoffs and randomly selects one Block-1 round for payment.

3.4 Block 2: structural change and reversal

After Block 1, pools are assigned to *recovery* or *persistence*. Under recovery, participants learn that Parliament has passed a budget, healthcare waiting times are falling, and public-service planning has resumed; the citizen-led multiplier rises from 1.50 to 2.50. Under persistence, they learn that the crisis continues and the multiplier remains 1.50. The leader-led multiplier, authority, and maximum transfer do not change. The transition screen reports both the conditions and applicable rates.

Participants report their expectations before the first Block-2 choice and again before the Round-10 choice, after nine rounds of experience. The first ballot measures immediate revision; the final ballot measures sustained reversal and is the primary endpoint. Participants answer the public-service and democratic-constraint questions only afterward. One Block-2 round is randomly paid, and both selected round payoffs are converted to euros and added to the participation payment.

4 Outcomes, Process Measures, and Estimands

Let $\mathbb{1}\{\cdot\}$ equal 1 when the condition inside braces is true and 0 otherwise. The final Block-1 ballot defines leader endorsement as

$$A_{i1} = \mathbb{1}\{V_{i,1,10} = \mathsf{A}\}. \tag{5}$$

For the theoretically relevant subgroup with $A_{i1} = 1$, democratic reversal is

$$R_i = \mathbb{1}\{V_{i,2,10} = \mathsf{D}\}. \tag{6}$$

The primary estimand is the average treatment effect of assignment to institutional recovery among Block-1 leader endorsers:

$$\tau_R = \mathbb{E}[R_i(1) - R_i(0) \mid A_{i1} = 1], \tag{7}$$

where $R_i(1)$ and $R_i(0)$ are participant i's potential reversal outcomes under assignment to recovery and persistence. The software delivers the assigned public-service conditions and corresponding fund rate to the entire pool, so the effect of assignment is the effect of that institutional environment. Moreover, A_{i1} is measured before assignment and is therefore a pretreatment characteristic. The primary test uses randomization inference that reproduces the pool-level assignment procedure. Regression estimates with pool-clustered standard errors can be reported.

Immediate reversal is

$$R_i^I = \mathbb{1}\{V_{i,2,1} = \mathsf{D}\} \quad \text{for } A_{i1} = 1. \tag{8}$$

Applying the same recovery-versus-persistence comparison to R_i^I estimates the immediate causal effect of recovery. The final-round outcome R_i remains the primary endpoint because it records whether reversal persists through the last ballot.

In simple terms

The primary estimand (τ_R) asks: among participants who choose “One person chooses for the group” in Block 1's final round ($V_{i,1,10} = \mathsf{A}$; $A_{i1} = 1$), how much more often does recovery ($T_g = 1$), rather than persistence ($T_g = 0$), lead them back to “Each person chooses” in Block 2's final round ($V_{i,2,10} = \mathsf{D}$; $R_i = 1$)? This is the average treatment effect of recovery for previous A supporters. Recovery improves the stated public-service conditions and equalizes productivity, while A still prevents free-riding.

Only one of $R_i(1)$ and $R_i(0)$ can be observed for each participant. Random assignment identifies τ_R by comparing reversal rates across conditions (T_g). The software delivers the assigned condition, so no separate measure of treatment compliance is required. Assignment and inference operate at pool level.

For each selected leader i, t_{igbr} is the required allocation and ρ_{igbr} the personal transfer in pool g, block b, and round r. A personal transfer is defined as the indicator $M_{igbr} = \mathbb{1}\{\rho_{igbr} > 0\}$, with intensity recorded in points and as $\rho_{igbr}/(5t_{igbr})$ when $t_{igbr} > 0$. For example, if the leader requires 10 points from each of five citizens, the fund contains $5t_{igbr} = 50$ points. Moving 5 points to the leader's payoff gives $M_{igbr} = 1$, $\rho_{igbr} = 5$, and $\rho_{igbr}/(5t_{igbr}) = 5/50 = 0.10$: a transfer occurred, its size was 5 points, and it represented 10% of the fund. Leader selection follows the balanced random rule described above, but transfers and participants' exposure to them depend on prior institutional choices and leader behavior. Their associations with subsequent votes therefore provide descriptive process evidence rather than separate causal effects of recovery.

Immediate reversal R_i^I records the first response to the new rates, before any Block-2 interaction (see Figure 5). The primary outcome R_i records the choice at the start of Round 10, after nine completed rounds of Block-2 experience. The measures can differ: a citizen may reverse immediately but not sustain that choice, or may return to D only later. Secondary outcomes include the change in leader-vote share across Rounds 8–10.

Participants report expected payoffs under both methods and the expected transfer, before the final Block-1 choice and the first and final Block-2 choices. These reports show whether assigned recovery changes perceived relative performance in the direction implied by the new citizen-led fund rate. They are secondary process measures. After the final ballot, a post-experimental questionnaire records perceived effectiveness and views on compulsory allocation, personal transfers, among others. These post-treatment measures show which considerations align with reversal; they provide descriptive process evidence, not causal mediation tests.

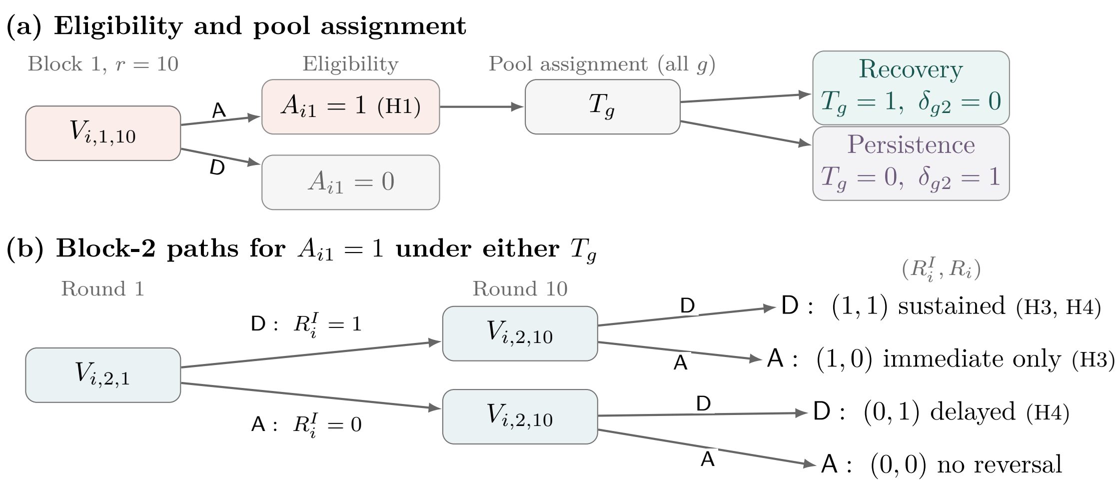


Figure 5: Decision tree for assignment and reversal. Each pool is assigned one value of T_g; the primary analysis follows participants with $A_{i1} = 1$.

Notes: The displayed ballots are $V_{i,1,10}$ (final Block 1), $V_{i,2,1}$ (first Block 2), and $V_{i,2,10}$ (final Block 2). The ordered pair (R_i^I, R_i) records the two Block-2 outcomes, with $1 =$ D and $0 =$ A. Thus, $(1,1)$ is sustained reversal; $(1,0)$ immediate only; $(0,1)$ delayed; and $(0,0)$ no reversal. Parentheses identify the corresponding hypotheses; H2 concerns transfers and is not shown.

5 Randomization, Sample, and Power

The sample comprises 400 participants in 40 ten-person pools. Treatment is assigned only after everyone finishes Block 1. Within each session, pools are ordered by their final-round Block-1 leader-vote share and paired. One pool in each pair receives recovery and the other persistence, at random; an unmatched pool is assigned by a fair draw. Pairing improves balance in the pretreatment number of participants eligible to reverse while preserving random assignment. A single visit removes panel attrition.

Power depends on three quantities: Block-1 endpoint support for A, the persistence-group reversal rate, and the similarity of outcomes within pools. Pilot estimates will inform pool-level simulations. Repeated rounds improve measurement and reveal trajectories; they do not turn 400 participants in 40 randomized pools into 8,000 independent observations. Production sessions require multiples of ten and, when feasible, an even number of pools. Each oTree session is a separate interaction and randomization cohort.

Strategic decisions do not time out automatically. The software records response time and flags decisions exceeding 90 seconds, but never generates a ballot or allocation. The laboratory manager monitors the session and prompts delayed participants to complete the current decision so the group can advance.

References

Dal Bó, Pedro, Andrew Foster, and Louis Putterman (2010). "Institutions and Behavior: Experimental Evidence on the Effects of Democracy." In: *American Economic Review* 100.5, pp. 2205–2229.

Fehr, Ernst and Simon Gächter (2000). "Cooperation and Punishment in Public Goods Experiments." In: *American Economic Review* 90.4, pp. 980–994.

Fischbacher, Urs and Simon Gächter (2010). "Social Preferences, Beliefs, and the Dynamics of Free Riding in Public Goods Experiments." In: *American Economic Review* 100.1, pp. 541–556.

Gächter, Simon, Elke Renner, and Martin Sefton (2008). "The Long-Run Benefits of Punishment." In: *Science* 322.5907, p. 1510.

Goodman, Sara Wallace (2021). "Immigration Threat, Partisanship, and Democratic Citizenship: Evidence from the US, UK, and Germany." In: *Comparative Political Studies* 54.11, pp. 2052–2083.

Gürerk, Özgür, Bernd Irlenbusch, and Bettina Rockenbach (2006). "The Competitive Advantage of Sanctioning Institutions." In: *Science* 312.5770, pp. 108–111.

Hamman, John R., Roberto A. Weber, and Jonathan Woon (2011). "An Experimental Investigation of Electoral Delegation and the Provision of Public Goods." In: *American Journal of Political Science* 55.4, pp. 738–752.

Hetherington, Marc J. and Elizabeth Suhay (2011). "Authoritarianism, Threat, and Americans' Support for the War on Terror." In: *American Journal of Political Science* 55.3, pp. 546–560.

Kingzette, Jonathan et al. (2021). "How Affective Polarization Undermines Support for Democratic Norms." In: *Public Opinion Quarterly* 85.2, pp. 663–677.

Kosfeld, Michael, Akira Okada, and Arno Riedl (2009). "Institution Formation in Public Goods Games." In: *American Economic Review* 99.4, pp. 1335–1355.

Markussen, Thomas, Louis Putterman, and Jean-Robert Tyran (2014). "Self-Organization for Collective Action: An Experimental Study of Voting on Sanction Regimes." In: *Review of Economic Studies* 81.1, pp. 301–324.

Ostrom, Elinor, James Walker, and Roy Gardner (1992). "Covenants with and without a Sword: Self-Governance Is Possible." In: *American Political Science Review* 86.2, pp. 404–417.

Saikkonen, Inga A.-L. and Henrik Serup Christensen (2023). "Guardians of Democracy or Passive Bystanders? A Conjoint Experiment on Elite Transgressions of Democratic Norms." In: *Political Research Quarterly* 76.1, pp. 127–142.

Sutter, Matthias, Stefan Haigner, and Martin G. Kocher (2010). “Choosing the Carrot or the Stick? Endogenous Institutional Choice in Social Dilemma Situations.” In: *Review of Economic Studies* 77.4, pp. 1540–1566.
Wunsch, Natasha, Marc S. Jacob, and Laurenz Derksen (2025). “The Demand Side of Democratic Backsliding: How Divergent Understandings of Democracy Shape Political Choice.” In: *British Journal of Political Science.*